

ELEC 328 Project Report
Assn 1 - Procedural Mesh Interactions
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I. Concept

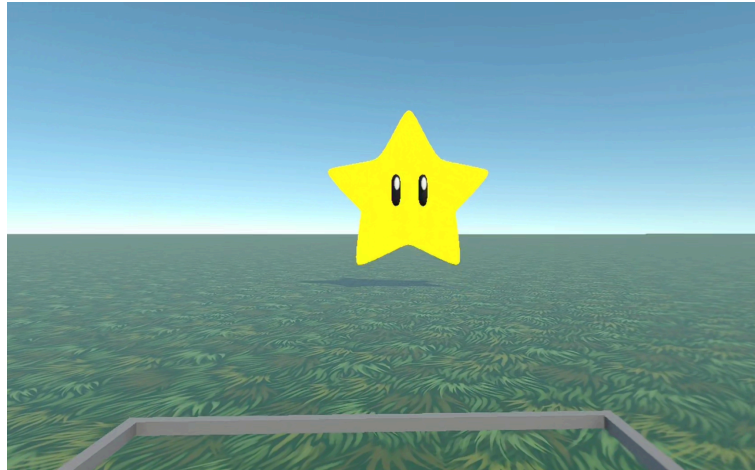


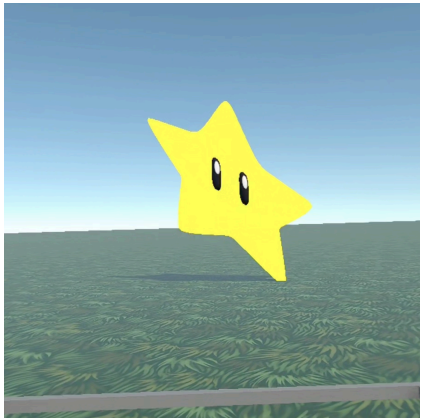
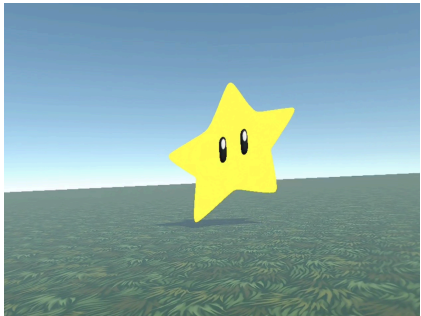
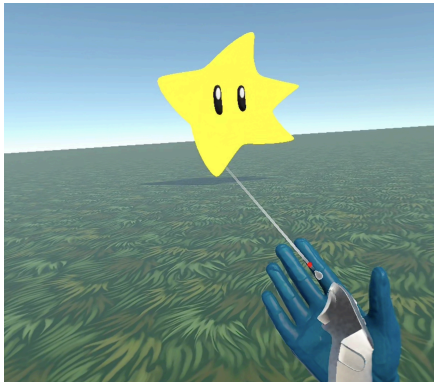
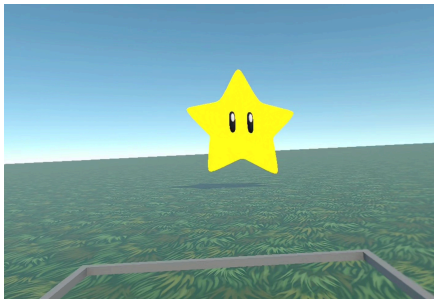
Figure 1. Mario Super Star Mesh

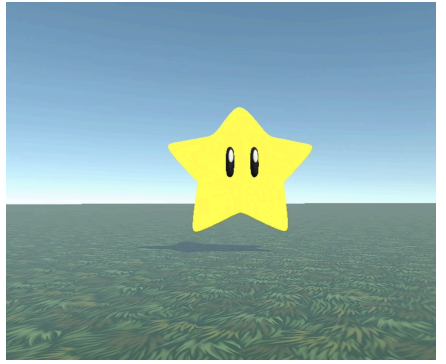
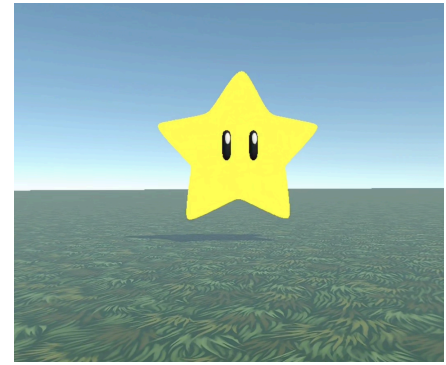
The objective of the assignment was to explore how mesh geometry can exhibit expressive, responsive behavior in a VR environment by manipulating it in real time. The project was implemented in the provided Reality Roost SDK scene to create a real time VR experience. I implemented three distinct deformations on a Mario Super Star mesh, including a continuous wave, a controller input twist, and a “nodding” bend on the upper half of the mesh. I learned how to modify mesh vertices, connect VR controller input to manipulate geometry, and design within the Reality Roost platform.

II. Technical Overview

The mesh deformations were implemented using a MonoBehaviour Script containing a simple finite state machine. The program includes three states corresponding to three deformations: Wave, Twist, and Nod. Controller inputs, namely the buttons A, B, and X, trigger transitions between each state. The first Wave deformation creates a continuous wave along the Y axis of the mesh by applying a sine function to the vertex positions over time using a for loop. The second Twist deformation rotates vertices around the Z axis using sine and cosine functions. It is input based; the left and right controller triggers control the direction and magnitude of the twist. Finally, the third Nod deformation rotates the upper portion of the mesh around the X axis (also using sine and cosine functions) and emphasizes the effect the farther the vertex’s distance from the center of the mesh.

III. Evidence of Requirements

Requirement	Where it is demonstrated	Screenshot
Continuous deformation	Experience video 0:07-0:15; Screenshot 1, 2	 Screenshot 1  Screenshot 2
Input-responsive deformation	Experience video 0:16-0:32; Screenshot 3	 Screenshot 3
Spatial placement (meshes outside the rails)	Experience video 0:09; Screenshot 4	 Screenshot 4

Third, visibly distinct deformation	Experience video 0:36-0:44; Screenshot 5, 6	 Screenshot 5  Screenshot 6
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IV. Assets, data, and citations

Item	Source / citation	How it was used
Super Mario Star	Source: SketchFab Author: gaelinix URL: https://sketchfab.com/3d-models/super-mario-star-cd988ff787734dec955cd76e8acdf327	The star was implemented as the mesh to be deformed in the assignment.
Stylized Grass Texture	Source: Unity Asset Store Author: LowlyPoly URL: https://assetstore.unity.com/packages/2d/textures-materials/glass/stylized-grass-texture-153153?srltid=AU7gw4ULDDTWMyb5WrzPhP5JmTW_2ZjgHFo82BeRQIDDEbXlaafewOoV	This item was used as the texture of the ground plane in the scene.

V. AI Use Disclosure

Tool	Used for	What was incorporated, changed, or rejected
ChatGPT	Unity scripting & debugging	I used ChatGPT to help develop and revise the mesh deformation script (input, wave, twist, nod). I specifically

		used it to test the program on my computer and then change the parameters to make it compatible with the VR headset.
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VI. Reflection

Overall, I was able to implement three distinct vertex-level deformations, one continuous and time based, one input based, and the third also continuous and time based. I was able to test the approaches using the Unity application on my computer and using the VR controllers in the Reality Roost. I learned how vertices can be manipulated in real time, and how the controller inputs can be used to vary between and control different behaviors (i.e. mesh deformations).

Initially, I tried an audio-based approach to the third deformation. I implemented a filter to detect the bass of the song, and then mapped the bass intensity to the vertex position such that the mesh “reacted” to the song. While it worked within the Unity program, the audio was not playing through the headset, despite multiple attempts to troubleshoot with Alex’s assistance. Next, I tried a proximity-based approach to the third deformation. While I got it to work, the program detected the proximity of the controllers based on an XR Origin (camera) game object, which interfered with the calibration of the Roost Rails. Removing the XR Origin caused the proximity to malfunction; hence, I changed the third deformation to an X-axis rotation as suggested by the TA Krutik.

To improve the project, I would further explore the approaches I tried previously but failed to implement, and clean up the behavior of the visible deformations.

VII. Honor Code Statement

On my honor, I have neither given nor received any unauthorized aid on this assignment.